

Knot-Causal Topology

Perspective, irreducibility, tangle substitution, and the completed-zeta sign

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Abstract. This paper gives the laboratory's knot intuition a typed mathematical carrier. An embedded knot, its received diagram, a braid presentation, a skein class, a prime connected-sum factorization, and an unknotting path are separated. Reidemeister, Alexander, Markov, and Schubert then become four distinct laws of presentation and sameness. A receiver produces crossings as transverse coincidences in a projected surface; movement through the projection discriminant produces Reidemeister events, while curvature belongs to the holonomy of alternate receiver transports rather than to a crossing count.

Open tangles provide the exact grammar sought for contextual substitution. Rational tangles supply a rigorous meeting of ratios, continued fractions, crossings, and boundary-relative topology. Skein relations linearize local alternatives without declaring them identical. This yields a precise definition of compression as substitution modulo every declared future context and receiver. It also explains why prime decomposition does not make transformation cost additive.

Applied to formulations of transcendental invariants and to the Riemann programme, this grammar sharpens rather than replaces the existing obligation. The critical half is typed as the fixed seam of completion and the half-density of the Mellin chart; it is not the positive member of a skein triple. A proof through knot-causal topology would require an exhaustive completed-zeta tangle category, a presentation-independent evaluation equal to the Weil response, and local substitutions which construct the already-missing natural contraction through an independently positive star representation. No such arithmetic skein carrier is asserted here.

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I The question and the result

The live question is not whether a visually knotted plot of the zeta function proves the Riemann Hypothesis. It is whether knot theory already contains exact structures for five ideas which the laboratory has repeatedly needed:

1. one underlying construction with many receiver-dependent crossing diagrams;
2. irreducible closed returns;
3. finite generators of presentation-equivalence;
4. boundary-preserving replacement of a local interior; and
5. invariants which survive those replacements without becoming the complete identity of the construction.

It does. The resulting translation is useful only when its levels remain separate:

Knot-theoretic object	Exact role	Holonic relation
Embedded knot	An embedding of a circle in an ambient three-body.	One closed causal return with an exterior.
Diagram	A generic two-dimensional projection with over/under data.	A receiver face carrying visible crossings.
Braid	An ordered presentation about a selected axis.	A causally ordered word whose closure forgets presentation data.
Tangle	An open embedding with a typed marked boundary.	A replaceable interior with an exposed interface.
Skein class	A linear quotient by declared local relations.	Receiver-exact substitution under a selected linear semantics.
Prime knot	A nonunit irreducible under connected sum.	An irreducible closed factor for one composition law.
Unknotting path	Crossing changes separated by isotopies.	A topology-changing trajectory, not a re-presentation.

The strongest new relation is between tangles and compression. An open interior may be replaced exactly when every declared exterior context and receiver gives the same consequence. This is the compositional form of receiver-exact compression. A skein relation is one linear method for proving such a substitution; it is not the assertion that its local alternatives are the same knot.

II Four laws of presentation and sameness

II.1 The knot is not the diagram

Definition 1 — Situated knot receiver and its crossing field Let $K : S^1 \rightarrow M^3$ be an oriented tame embedding in an oriented three-manifold. A **situated knot receiver**

$$\rho = (\Sigma_\rho, \pi_\rho, \nu_\rho)$$

consists of an oriented receiving surface, a generic projection $\pi_\rho : M^3 \rightarrow \Sigma_\rho$, and a coorientation or depth law ν_ρ sufficient to order the two preimages of every transverse double point.

Its crossing population is

$$\mathcal{C}_{\rho(K)} = \left\{ \{s, t\} : s \neq t, \pi_{\rho(K(s))} = \pi_{\rho(K(t))}, \text{ the projected tangents are transverse} \right\}.$$

Together with the source orientation, depth order, and local crossing sign, this population forms the received diagram $D_{\rho(K)}$.

The embedded knot K , its ambient-isotopy class $[K]$, its received diagram $D_{\rho(K)}$, and the scalar crossing count $|\mathcal{C}_{\rho(K)}|$ are four different objects. A crossing is a receiver-relative incidence between two source parameters. It is not an intrinsic vertex of the embedded knot.

Boundary. The receiver must be generic for the crossing population to consist of transverse double points. Tangencies, cusps, triple points, and failed depth order belong to the projection discriminant. Switching one over/under relation is a crossing change and may change knot type; it is not a Reidemeister re-presentation of the same knot.

The received crossing population can be written directly. For an oriented embedding $K : S^1 \rightarrow M^3$ and a generic receiver projection $\pi_\rho : M^3 \rightarrow \Sigma_\rho$,

$$\mathcal{C}_{\rho(K)} = \left\{ \{s, t\} : s \neq t, \pi_\rho K(s) = \pi_\rho K(t), (\pi_\rho K)'(s) \pitchfork (\pi_\rho K)'(t) \right\}.$$

Each member is an equality in the receiving surface with two distinct source parameters. Depth order says which branch is over. Orientation supplies a crossing sign. The embedded curve itself contains neither a literal self-intersection nor an intrinsic crossing vertex.

This resolves the apparent conflict between “crossings depend on perspective” and “knot type is invariant.” They refer to different objects. The projection may acquire or lose crossings while the ambient-isotopy class of K remains exact.

II.2 Reidemeister, Alexander, Markov, and Schubert

Theorem 2 — Diagram, braid, closure, and prime-factor sameness are generated by different laws For tame oriented links in S^3 :

1. two received link diagrams present ambient-isotopic links exactly when they are related by planar isotopy and finitely many Reidemeister moves;
2. every oriented link is the closure $\hat{\beta}$ of some braid $\beta \in B_n$;
3. two braids have ambient-isotopic oriented closures exactly when they are related by braid isotopy together with Markov stabilization and destabilization; in algebraic notation this includes conjugation $\beta \mapsto \alpha\beta\alpha^{-1}$ and $\beta \in B_n \mapsto \beta\sigma_n^{\pm 1} \in B_{n+1}$; and
4. under connected sum, oriented knot types form a commutative monoid whose unit is the unknot and whose nonunit elements factor uniquely, up to ordering, into prime knots.

Consequently a diagram, a braid word, a closed-link type, and a connected-sum factorization are presentations at different categorical levels. No move theorem at one level identifies the complete data at another.

Demonstration. The four clauses are the classical Reidemeister, Alexander, Markov, and Schubert theorems, respectively. Their conjunction gives the stated hierarchy. Reidemeister moves preserve a closure type while changing a diagram. Alexander’s theorem supplies braid presentations but not uniqueness. Markov’s theorem generates precisely the extra equivalence introduced by closure. Schubert’s theorem concerns the separate connected-sum monoid.

Boundary. A prime knot is irreducible under connected sum; it is not a knot which cannot be changed into the unknot by crossing changes. Every knot has a finite unknotting sequence, while a nontrivial knot cannot become the unknot through Reidemeister moves alone because those moves preserve its ambient-isotopy class. None of the four theorems makes crossing number, unknotting number, runtime, or another cost additive over prime factors.

The four classical theorems answer four different questions [1], [2], [3], [4], [5]:

Theorem	Question	Generators
Reidemeister	When do two diagrams present the same link?	Planar isotopy and the three local diagram moves.

Theorem	Question	Generators
Alexander	Can every oriented link be presented by an ordered braid?	Every link is the closure of at least one braid.
Markov	When do two braid presentations have the same closure?	Braid isotopy, conjugation, and stabilization/destabilization.
Schubert	How does a knot factor under connected sum?	A unique unordered multiset of prime-knot factors.

This is a precise refinement of the soul hierarchy. Reidemeister moves retain closure type while changing the diagram. Markov moves retain closure type while changing braid index and braid presentation. Prime decomposition retains the connected-sum object while exposing its factors. None preserves the complete causal path by which the particular occurrence was formed.

Alexander’s theorem is especially important for the laboratory’s axis language. It says that an oriented link can be re-presented as strands moving monotonically around a braid axis. The axis is part of that presentation; it is not an absolute coordinate embedded in the knot type. Markov’s theorem then states exactly how different axis-relative presentations can close to the same link.

II.3 A nontrivial knot cannot be isotoped into the unknot

The phrase “a nontrivial knot which cannot be unknotted” needs one operation specified.

1. Under ambient isotopy, every nontrivial knot remains nontrivial. This is what **nontrivial knot type** means. Reidemeister moves cannot turn its diagrams into an unknot diagram.
2. Under crossing change, every tame knot admits an unknotting sequence. Its unknotting number $u(K)$ is the minimum number of crossing changes over all diagrams and intervening isotopies.
3. An algorithm can decide whether a tame knot is the unknot; recognition is different from finding a minimal unknotting path [6].

A crossing change is not a new camera orientation. It passes one branch through another and changes the embedding type. The distinction is the topological counterpart of separating a receiver rechart from a causal deed.

II.4 Prime means irreducible under one composition

The unknot is the unit of connected sum. A prime knot is a nonunit K such that

$$K = K_1 \# K_2 \Rightarrow K_1 = 0_1 \text{ or } K_2 = 0_1.$$

Schubert’s theorem makes the connected-sum monoid a free commutative monoid on prime-knot types. This is a real structural analogy with positive integers under multiplication. It is not an identification:

Positive integer monoid	Oriented knot monoid in S^3
1 is the unit.	The unknot 0_1 is the unit.
Primes are multiplicatively irreducible.	Prime knots are connected-sum irreducible.
Factorization is a multiset of primes.	Factorization is a multiset of prime-knot types.
Arithmetic valuations record multiplicity.	Prime-knot multiplicities record connected-sum factors.

The analogy ends before cost, order, geometry, linking, or spectral response. The free generators belong to different monoids.

III Prime decomposition does not decompose transformation cost

Brittenham and Hermiller prove that unknotting number is not additive under connected sum [7]. For the $(2, 7)$ torus knot $K = 7_1$ and its mirror $|K|$,

$$u(K) = u(|K|) = 3, \quad u(K \# |K|) \leq 5 < 6.$$

The intermediate isotopies matter. A crossing change modifies the whole standing knot, another presentation is exposed, and the next useful crossing need not belong to either original summand. Unique factorization of objects therefore does not imply additive distance in the graph of topology-changing deeds.

This distinction is decisive for compression. A construction may have irreducible constituents while a joint contextual replacement is cheaper, more expensive, or simply incomparable with the sum of independent replacements. The complete boundary and peripheral relation matter. Even the knot group alone does not determine unknotting number; the knot together with its peripheral structure recovers the needed embedding information [7], [8].

IV Crossings form a receiver field

IV.1 Chambers and the projection discriminant

Theorem 3 — Receiver crossings are chamber data; curvature requires alternate transports Let $\mathcal{P}$ be a smooth parameter space of pairs (K, ρ) consisting of a tame embedding and a situated receiver, and let $\Delta \subset \mathcal{P}$ be the discriminant of nongeneric received diagrams. On each connected chamber of $\mathcal{P} - \Delta$, the crossing population, crossing signs, and depth order continue without a combinatorial event. A generic one-parameter path meeting a codimension one stratum of Δ produces the corresponding local Reidemeister event.

Suppose additionally that each receiver ρ carries a fiber E_ρ and each admitted receiver path γ carries transport U_γ . The ordered crossings along one path form a lineage, but they are not curvature. Curvature appears only when two alternate transports can be compared. On a discrete triangle,

$$\Omega_{012} = U_{02} - U_{12}U_{01},$$

or, for invertible edge maps, through the returned holonomy

$$H_{012} = U_{20}U_{12}U_{01}.$$

In a smooth connection chart this is the infinitesimal law

$$F_\nabla = dA + A \wedge A.$$

Demonstration. Generic transversality makes the received diagram stable under a small parameter variation until the path meets the discriminant. The local models for a generic codimension-one passage are precisely the Reidemeister diagram events. The second claim follows from the definition of a connection's curvature as the first-order path-dependence of parallel transport around an infinitesimal two-dimensional comparison cell. A single path has no alternate route with which to form that comparison.

Boundary. Crossing count can jump with receiver while knot type remains fixed. Conversely, a crossing change alters the over/under law and can change knot type without being a receiver motion. A nontrivial returned holonomy may disappear after quotienting to knot type while remaining visible to a framed, braid, metric, or causal-lineage receiver. A gyroparallelogram gives one specialized model of this failure of ordinary vector addition only when the transport law satisfies the gyrogroup axioms; it is not a universal curvature definition.

Let $\mathcal{P}$ parameterize the embedding and receiver together. The discriminant Δ contains nongeneric projections: tangencies, cusp events, triple coincidences, and other failures of the transverse-double-point

chart. Inside one chamber of $\mathcal{P} - \Delta$, the diagram continues without a combinatorial event. A generic one-parameter path through Δ gives a Reidemeister movie.

This is the rigorous form of visible crossings changing under precession. The receiver trajectory

$$\rho : I \rightarrow \mathcal{P}$$

does not sample unrelated images. It carries one ordered lineage of diagrams. The Reidemeister events are its discrete seams. Between seams, projected vertices and arcs move continuously.

Vassiliev's study of discriminant complements demonstrates the wider principle: knot invariants can arise from the topology of the space of maps after singular maps are organized as a discriminant [9].

IV.2 Three meanings of crossing

The word **crossing** now has three typed uses:

1. a projected double point in $D_{\rho(K)}$;
2. a crossing change, which changes the over/under relation and may change knot type; and
3. the crossing of two transformation paths in a comparison diagram.

Only the first is created by the receiving projection. Only the second is an unknotting deed. The third is categorical or geometric incidence and may exist without a knot. Treating the three as one scalar “number of crossings” would erase exactly the causality the laboratory is trying to retain.

IV.3 Curvature emerges from alternate receiver transports

A path $\rho(t)$ supplies order, winding, and discrete seam events. It does not by itself supply curvature. Curvature requires at least one alternate direction of comparison.

Let E_ρ be the receiver-relative fiber and $U_{ij} : E_i \rightarrow E_j$ the transport along an edge of a receiver triangle. Then

$$\Omega_{012} = U_{02} - U_{12}U_{01}$$

is an additive comparison residual, while

$$H_{012} = U_{20}U_{12}U_{01}$$

is the returned holonomy when the edge maps are invertible. In a smooth connection chart,

$$F_\nabla = dA + A \wedge A.$$

This is curvature from perspective over orders of time: two ordered receiver histories begin at the same cut, reach a comparable cut, and fail to induce the same transport. The receiver participates because its own motion supplies the paths being compared. No observer outside the system is needed.

At the coarse knot-type receiver, Reidemeister loops may have trivial holonomy because they return to the same ambient-isotopy class. A finer framed, braid, metric, tangle, or causal-lineage receiver can retain a nontrivial return. Curvature is therefore receiver-relative without becoming arbitrary.

IV.4 Where the gyroparallelogram fits

In a gyrocommutative gyrogroup, addition is associative only after the automorphism

$$\text{gyr}[a, b]c$$

corrects the comparison of alternate bracketings. In the Einstein gyrovector model this gyration is the algebraic carrier of Thomas rotation or precession [10]. A gyroparallelogram is consequently a concrete curved replacement for the Euclidean parallelogram law.

This is relevant when the receiver transitions genuinely compose by a gyrogroup law. Then $\text{gyr}[a, b]$ is a finite holonomy face of the noncommuting transports. It should not be installed as the universal law of every comparison complex. The general object is connection and holonomy; the gyroparallelogram is one exact hyperbolic specialization.

V Axis, unit, and ratio

Definition 4 — Typed axes, local units, and covariant ratios Let $\mathcal{D}$ be a collection of quantity types. A **typed axis** of type $d \in \mathcal{D}$ over a base B is an oriented real line bundle $A_d \rightarrow B$. A **unit** on a region $U \subset B$ is a nowhere-zero local section $u_d \in \Gamma(U, A_d)$. A quantity $q \in (A_d)_x$ has the coordinate $x_{u(q)}$ determined by

$$q = x_{u(q)} u_{d(x)}.$$

If $u'_d = au_d$ with $a : U \rightarrow \mathbb{R}^\times$, then

$$x_{u'}(q) = a^{-1} x_{u(q)}.$$

The quantity and its type remain; only its coordinate face changes.

Let $\Phi : A_d \rightarrow A_e$ be a typed linear comparison over the same base, and choose units u_d, u_e . Its scalar ratio face $r_{u_e \leftarrow u_d}$ is defined by

$$\Phi(u_d) = r_{u_e \leftarrow u_d} u_e.$$

Under $u'_d = au_d$ and $u'_e = bu_e$,

$$r_{u'_e \leftarrow u'_d} = \left(\frac{a}{b}\right) r_{u_e \leftarrow u_d}.$$

Thus the map Φ is primary and the displayed scalar is its receiver-dependent coordinate.

Independently, a bare rational number is the equivalence class

$$\frac{m}{n} = [m : n] \in \mathbb{P}^1(\mathbb{Q}), \quad n \neq 0,$$

under $(m, n) \sim (\lambda m, \lambda n)$ for nonzero $\lambda \in \mathbb{Q}$. It requires two entries but not two physical quantity axes. It becomes a rate, scale conversion, or measured ratio only after a typed comparison supplies that interpretation.

Boundary. Type, axis, unit, coordinate, and scalar ratio are different data. A unit is not merely a label: it is a selected nonzero frame in a typed one-dimensional fiber. A conversion between unlike types requires an actual constitutive map; dimensional labels alone do not create one. The glyph $\frac{1}{2}$ may denote a projective ratio, an affine midpoint, a group average, a half-density exponent, or a fixed seam. Those faces coincide only through declared maps.

V.1 An axis is typed transport space, not a drawn coordinate line

A typed axis is an oriented one-dimensional fiber or line bundle. Its type says what kind of comparison is admitted. Its orientation says which direction is positive. An origin, a unit, and a connection are additional choices.

If u is a local unit and $q = xu$, changing to $u' = au$ gives

$$x' = a^{-1}x.$$

The quantity has not changed merely because its coordinate has. This is the precise sense in which a unit is an addressable axis face: the label selects a local frame, while the complete quantity remains a typed geometric element.

A comparison between axes is a map

$$\Phi : A_d \rightarrow A_e.$$

Chosen units turn it into a scalar r by $\Phi(u_d) = ru_e$. Rebase either unit and the scalar changes covariantly. Ratios in measurement therefore do require comparable axes and a law between them. Bare arithmetic rational numbers do not: $\frac{m}{n}$ is already the projective class $[m : n]$.

This distinction preserves Brandon’s intuition while preventing it from becoming false by overextension:

1. a **rate** requires two typed variables;
2. a **scale factor** compares two frames;
3. a **probability quotient** compares a part to a conditioned total under a measure;
4. a **cross-ratio** compares four incidences projectively; and
5. a **rational number** is an algebraic equivalence class before any physical or informational interpretation.

V.2 The triangle face of one half

Let an oriented triangle have edge chain

$$\partial[0, 1, 2] = [1, 2] - [0, 2] + [0, 1].$$

If one receiver partitions its three edge occurrences into a distinguished base $B = \{e_0\}$ and a paired complement $L = \{e_1, e_2\}$, then

$$\frac{|B|}{|L|} = \frac{1}{2}.$$

This is a valid combinatorial face of one-per-two. It is not the ratio of the side lengths of an equilateral triangle, since those are $1 : 1 : 1$ after normalization. Direction and causal parity live in the oriented boundary signs and in whatever transport each edge carries.

The construction is nevertheless important. The smallest closed two-dimensional comparison has three oriented edges, alternate routes, and one boundary cancellation law. If those edges become hinges in a larger simplicial complex, their behavior is conditioned by the co-present faces; there is no context-free scalar state “per triangle.”

V.3 Why the critical half is not decimal convention

Write

$$s = \frac{1}{2} + z.$$

The completed involution $s \mapsto 1 - \bar{s}$ becomes

$$z \mapsto -\bar{z}.$$

The line $\text{Re}(s) = \frac{1}{2}$ is therefore the fixed locus of this reflection. It is also the square-root-density line for the standard multiplicative measure after the logarithmic/Mellin rebase. These are typed analytic roles, not a consequence of base ten.

The same glyph has other exact faces:

$[1 : 2]$	A projective rational coordinate.
$\frac{1}{2}$	The affine midpoint fixed by $x \mapsto 1 - x$.

$\frac{I+J}{2}$	The projector onto the fixed subspace of an involution J .
$\frac{\beta}{2}$	The half-density seam for $r^\beta d_r^r$.
$\text{Re}(s) = \frac{1}{2}$	The completed-zeta fixed seam when $\beta = 1$.

These are connected by specified constructions; they are not interchangeable because the numeral looks the same.

VI Tangles make interiors compositional

VI.1 An open knot has a boundary

A tangle is a properly embedded collection of arcs and circles in a three-ball or cylinder, with marked endpoints on the boundary. It is the natural unit of local substitution because an exterior can be held exact while the interior changes.

Definition 5 — Contextual tangle equivalence and receiver-exact substitution Fix an oriented marked boundary B . Let $\text{Tan}(B)$ be the collection of tangles with boundary B , modulo ambient isotopy relative to B . A compatible exterior context C composes with an interior tangle T by gluing, written $C[T]$.

For a declared compositional receiver family $\mathcal{R}$, define **contextual receiver equivalence**

$$T \underset{\mathcal{R}}{\approx} T' \quad \text{iff} \quad R(C[T]) = R(C[T'])$$

for every $R \in \mathcal{R}$ and every admitted compatible context C . Because contexts themselves compose, $\underset{\mathcal{R}}{\approx}$ is a congruence: an equivalent interior may be substituted inside every larger admitted exterior.

A receiver-exact tangle compression is a factorization through the quotient

$$q_{\mathcal{R}} : \text{Tan}(B) \rightarrow \text{Tan} \frac{B}{\underset{\mathcal{R}}{\approx}}.$$

It may discard distinctions between interiors only at the stated receiver and context boundary.

After linearizing over a ring R_0 , a skein relation is a local module relation such as

$$\alpha[T_+] + \beta[T_-] + \gamma[T_0] = 0.$$

It licenses linear replacement inside any compatible context in the resulting skein module. It does not assert $T_+ \approx T_-$, $T_+ \approx T_0$, or ambient isotopy of the three tangles.

Boundary. Equal closure under one receiver is weaker than contextual equivalence. Contextual equivalence is weaker than causal-soul equivalence unless the receiver family is jointly conservative. A skein quotient can identify distinct knots or retain torsion, and its coefficients may be signed or complex. Skein reduction is therefore not automatically positive, metric-decreasing, unique, or computationally cheaper.

The relation

$$T \underset{\mathcal{R}}{\approx} T' \quad \text{iff} \quad R(C[T]) = R(C[T'])$$

for every declared receiver R and compatible exterior context C is the rigorous form of “the same constituent may be used here.” It is not absolute identity. It is a congruence at a typed interface.

This also supplies a better definition of link substitution. The link is not cut at an arbitrary visual region. A boundary sphere or disk exposes a marked interface; a replacement tangle must meet exactly that interface; and the declared receiver determines which resulting consequences must remain exact.

VI.2 Rational tangles are ratios with crossing interiors

Conway’s rational tangles provide an unusually direct bridge between the laboratory’s ratio and crossing intuitions. Starting from the trivial tangles, a finite sequence of horizontal and vertical twists produces a rational tangle. Its twist word gives a continued fraction

$$[a_1, a_2, \dots, a_n] = a_1 + \frac{1}{a_2 + \frac{1}{\dots + \frac{1}{a_n}}}.$$

Kauffman and Lambropoulou prove that rational tangles are classified up to isotopy by their fraction in $\mathbb{Q} \cup \{\infty\}$ [11]. Thus different diagrams and flypes can carry the same boundary-relative tangle type, while the continued fraction records one exact generated path.

This is not evidence that every ratio is secretly a knot. It is an established domain in which:

1. a projective ratio;
2. a finite series of reciprocal rebases;
3. an oriented crossing word;
4. a boundary-relative embedding; and
5. an isotopy classification

belong to one construction.

A crossing change is the simplest rational subtangle replacement. Baker and Buck classify much wider rational subtangle replacements and show that their sites are constrained by the topology of the branched double cover [12]. Local replacement is therefore not free-form editing; the available replacements depend on the complete exterior.

VI.3 Skein relations linearize alternatives

Let L_+, L_-, L_0 agree outside one disk and differ inside it by positive crossing, negative crossing, and oriented smoothing. The Conway relation is

$$\nabla_{L_+}(z) - \nabla_{L_-}(z) = z \nabla_{L_0}(z).$$

The three links need not be isotopic. Their invariant values are linearly related. Przytycki’s skein module makes the construction general: start with the free module on ambient-isotopy classes and quotient by the submodule generated by declared local skein combinations [13].

This is why L_+ cannot be read as “the positive case” of the Weil criterion. The plus and minus name local oriented crossing species. The coefficients of a skein relation can be negative, Laurent, or complex. A skein recursion is not a positivity proof.

Jones’s planar-algebra framework supplies the more relevant positive refinement. Planar tangles act by multilinear contractions; an involution comes from reflection; and additional positivity hypotheses can turn the result into a star-algebraic/operator object [14]. Positivity is extra structure on the evaluation, not a gift of the strings.

VII Compression is contextual substitution

Theorem 6 — Exact local substitution is compression only through a compositional receiver Let $T \underset{\mathcal{R}}{\approx} T'$ be contextual receiver equivalence at boundary B . Then for every finite tower of admitted exterior contexts $C_{n[\dots C_1[-]\dots]}$ and every receiver $R \in \mathcal{R}$,

$$R(C_{n[\dots C_1[T]\dots]}) = R(C_{n[\dots C_1[T']\dots]}).$$

Thus replacing T by T' is an exact compression for that complete future context-and-receiver family.

If only one closure $C_0[T]$ and one receiver value agree, the same conclusion does not follow. If instead a skein relation $\alpha[T_+] + \beta[T_-] + \gamma[T_0] = 0$ is supplied, exact substitution holds linearly in the skein module but does not make any pair of the three interiors equivalent.

Demonstration. Contextual equivalence quantifies over every admitted context. Applying its defining equality first to C_1 , then regarding the result as an interior of C_2 , and continuing inductively proves the first claim. Equality under one context is not quantified over later contexts, so it cannot prove the converse. A skein relation is equality of a linear combination in a quotient module, not equality of its generators.

Boundary. The theorem does not supply a complete set of local relations, a terminating or confluent normal form, or a cost improvement. Exact contextual compression can be nonlocal to discover even when the replacement itself is local. Brittenham–Hermiller’s nonadditivity of unknotting number is a concrete warning that factorwise simplification costs need not add after composition.

An open computation, formula fragment, neural subnetwork, proof fragment, or geometric patch may be treated as a tangle only after its input/output boundary and composition law are supplied. Then an interior replacement is exact when every declared later context factors through the quotient.

This yields four separate questions:

1. **Equivalence:** do the two interiors have the same declared behavior in every admitted context?
2. **Presentation:** is that equivalence generated by a finite local move calculus?
3. **Normalization:** do the moves terminate or possess a canonical form?
4. **Cost:** is the replacement cheaper under the selected physical or computational measure?

None implies the next automatically. Reidemeister’s theorem answers the second question for knot diagrams while saying nothing simple about the shortest move sequence. Brittenham–Hermiller show that even a natural topology-changing cost is not additive over prime factors.

For learned networks, the implication is precise. We should not delete “weak” scalar weights or retain “strong” edges as if those numbers were context-free. A candidate subnetwork has an exposed activation interface. Its replacement is exact only for the ecology of contexts and receivers in which the two induced boundary maps agree. Approximate compression replaces equality by an explicitly chosen metric, measure, or test family; the residual must remain typed by that choice.

VIII Transcendental invariants as closures of formulation tangles

Pi, e , and other transcendental constants have many series, products, integrals, limits, recurrences, and algorithms. The common value is one closure receiver. The complete formulation is better represented as an open tangle with:

1. parameter and branch boundaries;
2. arithmetic coefficient currents;
3. an ordered generator or recurrence;
4. a convergence domain and discriminant;
5. an evaluation or limiting closure; and
6. alternate proven local substitutions.

For a formulation T and evaluation context C_{eval} ,

$$C_{\text{eval}[T]} = \pi$$

does not make all such T identical. Euler transformations, hypergeometric identities, contour deformations, continued-fraction contractions, and recurrence changes are skein-like only when a proved local identity makes them compositional on the stated domain.

This refines “the invariant can be attained from any perspective.” A lawful receiver can attain the invariant when its formulation has an admissible evaluation path. An arbitrary perspective with no decoder, convergence law, or transition map does not.

The knot carrier nevertheless adds something new to the formulation atlas. It distinguishes:

1. an **open formulation tangle**, whose interface can compose;
2. a **closed invariant face**, such as the returned value;
3. a **presentation move**, which changes the formula without changing the represented formulation;
4. a **skein substitution**, which linearly relates nonidentical formulations; and
5. a **prime factorization**, which applies only when a selected composition monoid has irreducibles.

The arithmetic populations of primes and semiprimes inside a series belong to the tangle interior. They can affect convergence, cancellation, monodromy, and available substitutions even when the closure value is the same.

IX Arithmetic topology and primitive returns

For $\text{Re}(s) > 1$,

$$\zeta(s) = \prod_p (1 - e^{-s \log p})^{-1},$$

$$-\zeta' \frac{s}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) e^{-sm \log p}.$$

This has the exact grammar of primitive closed returns: p labels a primitive period of length $\log p$ and p^m its m -fold traversal. Deninger develops the prime–knot–periodic-orbit analogy in this precise dynamical direction [15]. Connes and Consani construct an arithmetic-geometric instance in which periodic orbits have length $\log p$ and Frobenius appears as their monodromy [16], [17].

These results justify studying knot-like return structure around primes. They do not identify connected sum with integer multiplication, a knot polynomial with the Riemann zeta function, or diagram crossings with zeta zeros. The complete explicit formula remains population-level: prime powers on one side, zeros and archimedean terms on the other.

X What the knot grammar changes for RH

X.1 The Weil condition is not a choice of crossing sign

For the standard admitted test class, the Weil criterion asks for a Hermitian quadratic response

$$Q_{W(f)} \geq 0$$

for every test current, and this positivity is equivalent to RH. When the zeros lie on the critical line, the zero contribution can be represented in the required square-norm form. An off-line zero produces a genuine negative direction for a suitable test. Invertible receiver recharting cannot change that inertia.

This is unrelated to calling one member of a skein triple L_+ . The latter is an oriented local crossing. The former is positivity of a global Hermitian form over an exhaustive test space.

X.2 The fixed half has an orbit interpretation

Center $s = \frac{1}{2} + z$ and attach a return of length ℓ :

$$\lambda_{\ell(z)} = e^{-\ell z}.$$

Completion gives

$$\lambda_{\ell(-\bar{z})} = \frac{1}{\lambda_{\ell(z)}}.$$

Therefore

$$\operatorname{Re}(z) = 0 \quad \leftrightarrow \quad |\lambda_{\ell(z)}| = 1.$$

The critical seam is the unit-modulus locus of the centered return. The functional equation pairs inside and outside radial sheets but does not force every zero orbit to be fixed. The figure-eight knot supplies the finite-dimensional counterimage: reciprocal Alexander monodromy can have one eigenvalue inside and one outside the unit circle.

In orbit language, completion partitions a zero population into:

1. fixed orbits on $\operatorname{Re}(s) = \frac{1}{2}$; and
2. paired size-two orbits off the line.

RH says the nontrivial zero population consists only of fixed orbits. The existence of the involution proves the partition, not the absence of the second species.

X.3 The exact skein-shaped proof architecture

Corollary 7 — An exhaustive positive completed-zeta skein carrier would imply RH Let $\mathcal{Z}$ be a boundary-typed involutive linear tangle category whose objects and morphisms present the complete completed-zeta test currents, prime admissions, support changes, archimedean returns, and receiver recharts. Suppose:

1. proved local relations give a presentation-independent and exhaustive quotient of those currents;
2. an involution-preserving evaluation functor sends gluing to operator composition and closure to the complete Weil quadratic response;
3. each generating local relation supplies, independently of the desired sign, the natural contractive filler between the established amplitude functors X and Y ; and
4. these local fillers are compatible with every relation, so they descend to one natural transformation Γ on the quotient.

Then

$$Y_j = \Gamma_j X_j, \quad \|\Gamma_j\| \leq 1$$

for every completed test object j . Hence

$$Q_{W(f)} = \|X_j f\|^2 - \|Y_j f\|^2 \geq 0$$

on the exhaustive Weil test space, and the Riemann Hypothesis follows by the Weil criterion.

Demonstration. Presentation independence makes the component assigned to a current independent of its braid, diagram, or reduction path. Compatibility with gluing and receiver arrows makes the components natural. The assumed local contractivity therefore gives the natural contraction required by the natural-contraction RH corollary, which yields Weil positivity and RH.

Boundary. No such completed-zeta skein category or positive evaluation functor has been constructed here. Ordinary Reidemeister, Markov, Alexander, Conway, Jones, HOMFLY, or Kauffman relations do not satisfy the stated zeta trace identity by default. A standard skein relation may have signed or complex coefficients and therefore supplies no positivity on its own. The missing object remains the arithmetic–archimedean natural contraction, now with a possible local presentation grammar.

Knot theory now supplies a possible presentation grammar for the existing RH obligation. Let $\mathcal{Z}$ be a proposed completed-zeta tangle category. Its boundaries would type:

1. finite-prime apertures;
2. archimedean Fourier–Poisson–Mellin transport;
3. support growth;
4. test-current involution; and
5. receiver recharting.

Its local relations would have to be proved analytic substitutions: Euler admission, Poisson summation, Mellin rebase, support continuation, or another actual identity. A closure evaluation

$$E : \mathcal{Z} \rightarrow \text{Herm}$$

would have to equal the complete Weil response, not merely resemble it. Reidemeister/Markov analogues would prove that E is independent of presentation. A positive star representation would then have to turn every generating substitution into the components of the already-required natural contraction

$$Y_j = \Gamma_j X_j, \quad \|\Gamma_j\| \leq 1.$$

If all of that were constructed exhaustively, RH would follow. This is a conditional proof architecture, not a proof.

X.4 The genuinely new obstruction exposed by skein theory

Skein recursion alone is insufficient because ordinary skein coefficients are not positivity-preserving. A local identity of the form

$$\alpha E(L_+) + \beta E(L_-) + \gamma E(L_0) = 0$$

can have signed or complex coefficients. Repeated resolution may compute an invariant while alternating signs. It need not preserve the positive semidefinite cone.

The proof-bearing local object is therefore more specific:

> an arithmetic–archimedean tangle relation whose evaluation is a > completely positive or contractive star-compatible map, whose closure is > exactly the Weil form, and whose relations are coherent under every > prime/support/receiver gluing.

This does not rename the old wall. It identifies a possible **generator grammar** for constructing the missing natural contraction and states the additional property that ordinary skein relations lack.

Convexity enters only after evaluation: positive semidefinite forms and operator contractions form convex sets. Concavity may enter through a log-determinant barrier on a positive-definite finite restriction. Neither is a topological invariant of a knot diagram.

XI Compression and RH share one local-to-global problem

The machine-learning and RH applications now meet at a real mathematical relation rather than a metaphor.

Structure	Compression specialization	RH specialization
Boundary	Activation, state, or callable interface.	Test-current, support, prime, and archimedean interface.
Interior	A contextual subnetwork or algorithmic constituent.	One arithmetic–archimedean formulation tangle.
Context	The ecology of later uses.	Every admissible Weil test and aperture continuation.
Receiver	Declared behavior, metric, or task family.	The complete Weil quadratic response.
Substitution	A boundary-equivalent replacement.	A proved local analytic relation.
Positive law	An explicit approximation/error geometry when needed.	A natural contraction or completely positive evaluation.

Both applications fail if one equal output is mistaken for contextual equivalence. Both fail if an irreducible decomposition is mistaken for an additive simplification cost. Both require local substitutions to remain coherent when embedded in larger exteriors.

The difference is jurisdiction. Compression may select a bounded receiver family and tolerate a declared residual. RH quantifies over the complete admissible Weil test space and tolerates no unaccounted negative direction.

XII Derived vocabulary

The requested terms can now be stated without metaphor:

- **Axis:** a typed oriented one-dimensional transport fiber; origin, unit, and connection are additional local data.
- **Unit:** a chosen nonzero local frame in that fiber. The label addresses the frame; rebase changes coordinates covariantly.
- **Ratio:** algebraically, a projective class $[m : n]$; operationally, the scalar coordinate of a declared comparison between typed axes.
- **Crossing:** for a knot receiver, a transverse coincidence of two distinct source parameters in the received surface, with depth and orientation data.
- **Knot invariant:** a receiver which factors through ambient-isotopy class. It is not necessarily complete.
- **Prime knot:** a nonunit irreducible under connected sum.
- **Unknotting:** a topology-changing path through crossing changes and intervening isotopies, not isotopy alone.
- **Tangle:** an open embedded interior with a typed marked boundary.
- **Skein relation:** a local linear relation among nonidentical tangle or link alternatives.
- **Compression:** contextual receiver equivalence followed by a quotient or representative choice.
- **Perspective curvature:** the holonomy or comparison residual between alternate receiver transports over a two-dimensional parameter cell.

XIII Present boundary

This derivation changes the programme in three concrete ways.

First, it supplies a standard topological carrier for the laboratory’s relative crossing field. Second, it replaces vague “link substitution” with typed tangle gluing and contextual congruence. Third, it identifies a new local condition for an RH-directed construction: the desired arithmetic–archimedean relations

must preserve positivity through a star-compatible evaluation, not merely compute the same invariant by a skein recursion.

It does not prove RH. The same global sign wall remains: construct the completed-zeta natural contraction, or an equivalent positive trace carrier, from the actual arithmetic and archimedean relations. Knot theory now suggests how that carrier could be presented and checked for independence of local formulation. It does not supply its arithmetic generator.

The most informative next proof-directed object is therefore not another knot visualization or a census of prime knots. It is one exact local Poisson–Mellin/Euler-support substitution square, written as an open tangle, followed by a test of whether its operator evaluation is star-compatible and contractive on the already-derived dyadic character cell. A positive result would construct the first generator of the proposed arithmetic skein carrier; a negative result would rule out this route without moving the wall again.

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